

Physics-Informed Multi-Scale Adaptive Graph Learning for Unified Event Detection, Localization, and Classification in Power System

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Abstract—This article proposes a physics-informed multi-scale adaptive graph neural network (PI-MSA) for rapid event detection, localization, and classification (ED-L-C) of single and multi-source coupling events in power systems to prevent large-scale failures. Prior work decomposes ED-L-C tasks into separate classification subtasks. Instead, this work frames ED-L-C as a unified regression problem through a spatial-aware and task-coupled three-dimensional target formulation, enabling one-shot inference of all abnormal events and their class probabilities at both node and edge levels. By incorporating physical grid topology and shortest-path admittance encoding through parallel multi-scale networks, this unified one-stage pipeline enables joint optimization and learning of shared graph features as well as progressive inter-task dependencies. Experimental validation on five power systems of varying scales and configurations demonstrates that PI-MSA achieves over 75% mean average precision and 90% localization accuracy across various scenarios, outperforming traditional methods by over 40% and exceeding state-of-the-art methods by at least 5%. The model maintains robust and accurate event analysis under topology changes and reduced observability conditions while achieving computational efficiency at 225 frames per second, enabling real-time monitoring and analysis for modern power systems.

Index Terms—Event detection localization and classification, modern power system, multi-source events, multi-scale adaptive graph network, physics-informed.

I. INTRODUCTION

A. Background and Motivation

MODERN power systems face a structural evolution driven by the integration of renewable energy sources (RES), harmonic load, and smart grid technologies, which heighten operational variability and uncertainty. This increased complexity renders the grid more vulnerable to disturbances and

introduces emerging operational risks. As a result, power system events such as temporary faults, line trips, load fluctuations, and cyber-physical incidents are becoming more frequent and diverse [1], [2]. These events can propagate rapidly across interconnected networks, potentially leading to grid failures and widespread system disruptions [3]. As demonstrated in Table I, multi-source events worldwide have precipitated significant blackouts and economic losses in recent years. Many of these major blackouts stem from minor incidents that were exacerbated by delayed and inaccurate event analysis. To enable timely intervention, rapid and accurate event detection [4], localization [5], and classification (ED-L-C) [6] capabilities has become crucial for maintaining system stability and preventing catastrophic failures in modern power grids [7], [8].

Recent advances in measurement technologies, particularly the widespread deployment of distribution-level phasor measurement units (D-PMUs), have enhanced system observability [9], [10], [11]. These devices sample at 30–120 Hz, capturing system dynamics with high temporal resolution through synchronized measurements. This rich data stream theoretically enables more accurate ED-L-C analysis [12], [13], [14]. However, extracting actionable insights from these high-dimensional, heterogeneous data poses significant computational and methodological challenges, necessitating an efficient and unified framework capable of real-time multi-source complex event analysis in increasingly dynamic power systems [15].

B. Literature Review and Technical Challenges

To address the analytical challenges, extensive research efforts have been devoted to ED-L-C tasks in recent years [6], [13], [16]. The methodology for ED-L-C has evolved from physics-based models to data-driven approaches, and more recently toward physics-informed data-driven frameworks, driven by the increasing complexity of modern grids and the availability of high-resolution measurement data from D-PMUs.

Physics-based ED-L-C methods are built upon fundamental physical laws of power systems and achieve event analysis through the establishment of mathematical models that describe electromagnetic transient processes [17]. The modeling process typically involves system parameter identification, equivalent circuit modeling, and differential equation solving, followed by time-frequency domain analysis of electrical quantities to

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TABLE I
NOTABLE MAJOR POWER SYSTEM ACCIDENTS TRIGGERED BY MULTI-SOURCE EVENTS (2021–2025) *

Year	Location	Multi-source Event Description	Impact Scope and Estimated Economic Loss
2025	Spain and Portugal	Preliminary findings indicating France-Catalonia 400 kV line fault caused European grid separation and cascading outages.	Cross-border transmission fault; over 50 million Iberian Peninsula residents affected; traffic disruptions due to signal failures.
2025	Argentina	High-voltage power line failures during an extreme heatwave led to simultaneous trips of multiple substations and two massive blackouts	Over 622 000 customers (≈ 2 million people) left without power; widespread transport, commercial, and government service disruptions;
2024	Oregon, USA	Extreme ice and snow weather caused multiple distribution substation failures and distribution line damage.	Approximately 100 000 people affected; multiple distribution substations damaged; millions of dollars in losses.
2024	Texas, USA	Hurricane with extreme heat, multiple distribution lines, and infrastructure damaged causing large-scale distribution system failures.	2.7 million customers affected; numerous distribution poles collapsed; some areas without power for over a week.
2022	Puerto Rico	Hurricane Fiona caused simultaneous damage to multiple transmission and distribution nodes, triggering cascading failures.	Complete island outage; millions affected; estimated \$1 billion USD in infrastructure damage and business interruption.
2021	Germany	Extreme rainfall and catastrophic flooding caused severe damage to multiple distribution equipment and medium-low voltage lines.	Over 200 000 customers without power; some areas took over 6 months to restore power; estimated loss exceeding €46 billion.

* The root causes and event descriptions for 2025 incidents are based on preliminary reports and publicly available information as of manuscript preparation. Final investigation findings may provide additional details.

support ED-L-C tasks. While these methods provide strong theoretical foundations and interpretability, their dependence on precise system parameters and simplified assumptions limits their effectiveness in modern distribution networks with high renewable penetration and dynamic loads.

To address the limitations of physics-based methods in complex modern grids, data-driven methods have emerged as promising alternatives for ED-L-C tasks, with deep learning techniques showing particular effectiveness in complex pattern recognition. These methods employ various network architectures such as convolutional neural networks, recurrent neural networks, or attention-based Transformer networks [18], [19], combined with supervised [20], [21], weakly-supervised [22], semi-supervised [23], or unsupervised learning [24] strategies according to data characteristics and task requirements. By learning effective feature representations from event data, these approaches demonstrate superior performance in event analysis. However, data-driven methods depend on high-quality historical data and offer limited physical interpretability, raising reliability concerns in critical decision-making scenarios.

To leverage the complementary strengths of both paradigms while addressing their respective limitations, physics-informed data-driven methods have emerged as a promising research direction that integrates domain knowledge into learning frameworks to ensure both event analysis accuracy and physical consistency [25]. These approaches primarily operate through three mechanisms: constraint-based methods that incorporate physical laws such as Kirchhoff's laws and power balance equations as regularization terms in loss functions [26], architecture-informed designs that embed topological and physical priors directly into network structures such as graph neural networks [27], [28], [29], [30], and knowledge-guided learning frameworks that utilize domain expertise such as impedance characteristics and power equipment parameters to direct feature extraction and model training processes [31]. These hybrid approaches have demonstrated improved generalization capabilities and reduced data requirements compared to purely data-driven methods. However, they still face significant challenges in balancing the trade-offs between physical constraints and learning flexibility. Specifically, three open challenges remain: (1) most ED-L-C frameworks treat subtasks independently, leading to error accumulation in sequential processing or insufficient inter-subtask dependency modeling, leaving room for unified formulations;

(2) jointly modeling multi-scale spatial dependencies and temporal dynamics under complex scenarios remains non-trivial; (3) further exploiting the synergy between circuit-theoretic principles and graph-structured learning could enhance interpretability and physical consistency in safety-critical scenarios.

C. Contributions

To address the technical challenges above, this article proposes a physics-informed multi-scale adaptive (PI-MSA) graph neural network with a novel ED-L-C framework for enhanced power system event analysis. Rather than treating the three subtasks independently, the proposed framework unifies ED-L-C into a one-stage regression problem via a spatial-aware three-dimensional (3-D) output tensor, enabling efficient and accurate multi-source event analysis under complex scenarios. The main contributions of this article are summarized as follows.

- 1) This study develops a PI-MSA graph neural network for precise spatial-temporal feature learning from complex multi-source event patterns. The method employs parallel architectures where physical grid topology captures local spatial dependencies and shortest-path admittance encoding extracts global spatial relationships, enabling adaptive modeling of multi-scale dynamic interactions between system nodes.
- 2) This study introduces a spatial-aware and task-coupled 3-D output structure that frames ED-L-C as a unified one-stage regression problem, where spatial indices inherently encode event locations and classification is structurally conditioned on detection confidence. This formulation enables one-shot inference for all spatially separated abnormal events and their class probabilities at both node and edge levels. The framework eliminates the need for defining new classes for coupled waveform patterns and avoids error accumulation between subtasks.
- 3) The proposed model and ED-L-C framework is systematically evaluated on five test systems with varying scales and voltage levels. Comparative studies demonstrate the effectiveness of PI-MSA in processing complex multi-source events, while robustness and generalization analysis validate its adaptability under diverse operating conditions and grid configurations. Ablation studies further verify the contribution of each designed module to the overall performance.

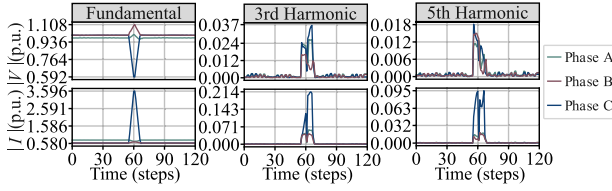


Fig. 1. H-PMU measurements for a C-G fault event at fundamental phasor versus at third and fifth harmonic phasor measurements.

The rest of this article is organized as follows. Section II presents a description of ED-L-C problems. Section III describes the details of the proposed PI-MSA model. Section IV outlines the training and inference phase. Section V introduces the experimental setup. Section VI presents and discusses the obtained results from different case studies. Finally, Section VII concludes this article.

II. PRELIMINARIES AND PROBLEM FORMULATION

This section presents basic definitions, the challenges posed by multi-source event scenarios, and the problem formulation for power system multi-source events spatial-temporal analysis.

A. Basic Definitions and Assumptions

Definition 1 (Power Grid Graph): Let $G = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ represent the power grid, where $\mathcal{V} = \{v_1, \dots, v_n\}$ denotes the set of nodes and $|\mathcal{V}| = N$. $\mathcal{E} = \{e_1, \dots, e_m\}$ represents the set of edges and $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the adjacency matrix.

Definition 2 (Graph Signal Matrix): At each time step t , the D-PMU measurements across all nodes form a graph signal $\mathbf{X}_t = (\mathbf{x}_{1,t}, \mathbf{x}_{2,t}, \dots, \mathbf{x}_{n,t}) \in \mathbb{R}^{N \times C}$, where C is the number of feature channels. For all nodes across the full-time window, $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_T]^\top \in \mathbb{R}^{N \times C \times T}$ is represented as the graph signal feature matrix, where T is the time window length. In this study, features include voltage magnitudes V_ϕ^{mag} , current magnitudes I_ϕ^{mag} , and power factors PF_ϕ for each phase $\phi \in \{A, B, C\}$, with 120 Hz reporting rate.

Assumption 1 (Harmonic Phasor Measurement): In this study, D-PMUs are assumed to be extended to harmonic phasor measurement units (H-PMUs) [5], which can selectively measure odd harmonics in addition to the fundamental frequency, as illustrated in Fig. 1. When higher-order harmonics are considered, the measured phasors are concatenated as follows:

$$\mathbf{X}_{\text{har}} = [\mathbf{X}_{\text{har},1} \parallel \mathbf{X}_{\text{har},3} \parallel \mathbf{X}_{\text{har},5}], \quad (1)$$

where $\mathbf{X}_{\text{har}} \in \mathbb{R}^{N \times 3C \times T}$ represents the concatenation matrix of the odd harmonics, $\mathbf{X}_{\text{har},1}$, $\mathbf{X}_{\text{har},3}$, and $\mathbf{X}_{\text{har},5}$ denote the fundamental, third, and fifth harmonic phasor measurements captured by the H-PMUs, respectively.

B. Multi-Source Events and Challenges

Fig. 2 presents an example scenario with eight-sources event captured from nine spatially distributed D-PMUs in the IEEE 34-node system. Fig. 2(a) shows the spatial locations and temporal intervals of each event, with color coding corresponding

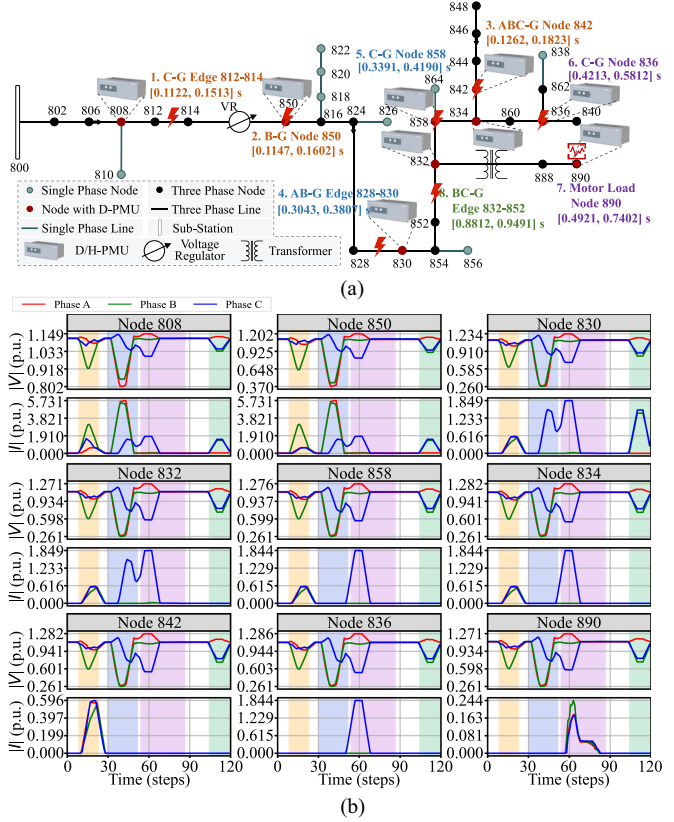


Fig. 2. (a) Spatial locations and temporal intervals of eight event sources in the IEEE 34-node system. (b) Three-phase voltage and current waveforms from 9 D-PMUs under an eight-sources event scenario. Background colors indicate event coupling levels: isolated (green), overlapping (purple), coupled (blue), and highly coupled (orange).

to the waveform segments in Fig. 2(b). For instance, events 1, 2, and 3 (orange) in Fig. 2(a) correspond to the orange-shaded waveform segments on the left side of Fig. 2(b). As shown in Fig. 2(b), the measured waveforms within this observation window exhibit complex patterns due to multiple event sources. The key challenges are summarized as follows.

- 1) **Variable Event Sources:** The number of event sources varies across samples and is unknown *a priori*, requiring the model to autonomously detect, localize, and classify all events.
- 2) **Temporal Coupling:** Events overlap in time (e.g., Events 1–3 occur nearly simultaneously around 0.11–0.18 s), causing superimposed signatures that evolve dynamically as events start and end throughout the observation window.
- 3) **Spatial Propagation:** Each event propagates beyond its source location, affecting upstream and downstream buses. Moreover, events occur at both node level and edge level, requiring simultaneous ED-L-C at two topological levels.
- 4) **Same-Class Discrimination:** This example contains three C-G faults (Events 1, 5, 6) at different locations, which must be identified as separate sources rather than a single event.

These challenges, combined with the requirement for real-time situational awareness, necessitate the development of an effective graph feature learning and unified framework that jointly addresses ED-L-C task for complex multi-source scenarios.

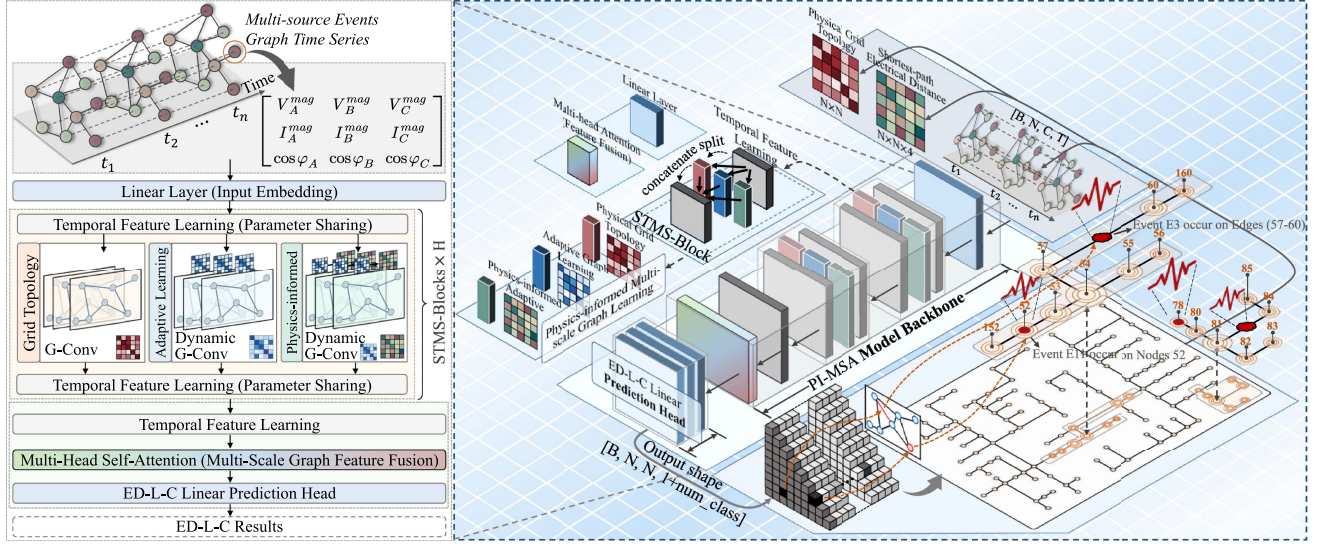


Fig. 3. Overview of the PI-MSA framework and developed multi-source event detection, localization, and classification framework.

C. Problem Statement

For a power grid G with graph signal matrix $\mathbf{X}$, the multi-source ED-L-C task can be viewed as a joint learning problem that integrates graph signal measurements with power system physical information to analyze anomalous events at both node and edge levels over graph time series, formulated as:

$$\mathcal{F} : (G, \mathbf{X}, \mathbf{Z}) \rightarrow (E_{\text{num}}, \{ (E_{\text{det}}^i, E_{\text{loc}}^i, E_{\text{cls}}^i) \}_{i=1}^{E_{\text{num}}}), \quad (2)$$

where $\mathbf{Z}$ is the physical parameter matrix of power system. $E_{\text{num}} \in [0, E_{\text{num}}^{\text{max}}]$ denotes the number of detected active events: normal operation ($E_{\text{num}} = 0$), single-source event ($E_{\text{num}} = 1$), multi-source event ($E_{\text{num}} \in [2, E_{\text{num}}^{\text{max}}]$) with multiple concurrent events. E_{det}^i , E_{loc}^i and E_{cls}^i indicate the event detection, localization, and classification results for the i -th event source among the current E_{num} events, respectively.

III. METHODOLOGY

A. Overview of Proposed PI-MSA Framework

As illustrated in Fig. 3, PI-MSA is built on a spatial-temporal multi-scale learning block (STMS-Block) with a temporal-spatial-temporal architecture, processing input features through three stages: (1) a shared temporal layer captures evolving dynamics across all nodes; (2) three parallel graph convolution pathways extract spatial dependencies using physical grid topology, adaptive adjacency, and a physics-informed shortest-path admittance matrix; (3) each pathway applies independent temporal learning, followed by multi-head attention for multi-scale feature aggregation and an ED-L-C prediction head. Multiple STMS-Blocks are stacked to enable hierarchical extraction of both fine-grained local event signatures and coarse-grained global system patterns.

B. Temporal Feature Learning in Power Systems

Temporal feature learning captures the evolving dynamics of power system events. We employ temporal gated convolution

(TGC) networks, which combine causal convolution with gated linear units to model the complex temporal patterns inherent in multi-source scenarios. Given the temporal features $\mathbf{x}_i^{(\ell, l)} \in \mathbb{R}^{T \times C_\ell}$ of node i at the l -th TGC layer in the ℓ -th STMS-Block, the causal convolution can be expressed as:

$$\mathbf{x}_{i,1}^{\text{Tconv},(\ell, l)} = \mathcal{T}_1(\Gamma_1 * \mathbf{x}_i^{(\ell, l)}), \mathbf{x}_{i,2}^{\text{Tconv},(\ell, l)} = \mathcal{T}_2(\Gamma_2 * \mathbf{x}_i^{(\ell, l)}), \quad (3)$$

where $\Gamma_1, \Gamma_2 \in \mathbb{R}^{K \times C_\ell \times C_T}$ are learnable temporal convolution kernels, C_ℓ and C_T denotes the input and output feature channels, K is the 1-D causal convolution kernel size, and $\mathcal{T}_1(\cdot)$, $\mathcal{T}_2(\cdot)$ are parallel causal convolution operations ensuring no future information is used. Each convolution reduces the temporal length by $K - 1$ without padding. The output $\mathbf{x}_{i,1/2}^{\text{Tconv},(\ell, l)} \in \mathbb{R}^{(T-K+1) \times C_T}$ are then processed through a gating unit:

$$\mathbf{x}_i^{(\ell, l+1)} = \sigma_{\text{sigmoid}}(\mathbf{x}_{i,1}^{\text{Tconv},(\ell, l)}) \odot (\mathbf{x}_{i,2}^{\text{Tconv},(\ell, l)} + \mathbf{x}_i^{(\ell, l)}), \quad (4)$$

where $\sigma_{\text{sigmoid}}(\cdot)$ denotes the sigmoid function, $\odot$ represents the Hadamard product. The gating unit controls feature importance, while the residual connection facilitates training.

C. Spatial Feature Learning in Power Systems

Event propagation in power systems exhibits strong spatial dependencies governed by physical connectivity and electrical characteristics between buses. To capture these, we employ Chebyshev graph convolution on the normalized Laplacian matrix $\mathbf{L} = \mathbf{I}_N - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}$, where $\mathbf{D} \in \mathbb{R}^{N \times N}$ is the degree matrix with $D_{ii} = \sum_j A_{ij}$. Given the spatial input features $\mathbf{X}_t^{(\ell)} \in \mathbb{R}^{N \times C_T}$ at time t after temporal processing, the graph convolution in the ℓ -th STMS-Block can be expressed as:

$$\mathbf{g}_t^{(\ell)} = \sum_{k=0}^{K-1} \theta_k^{(\ell)} \mathcal{P}_k(\tilde{\mathbf{L}}) \mathbf{X}_t^{(\ell)} + \mathbf{b}^{(\ell)}, \quad (5)$$

where $\theta_k^{(\ell)} \in \mathbb{R}^{C_T \times C_S}$ are trainable polynomial coefficients, $\mathbf{b}^{(\ell)} \in \mathbb{R}^{C_S}$ is the learnable bias term, and $\mathcal{P}_k(x)$ are Chebyshev polynomials computed recursively as $\mathcal{P}_0(x) = 1$, $\mathcal{P}_1(x) = x$,

$\mathcal{P}_k(x) = 2x\mathcal{P}_{k-1}(x) - \mathcal{P}_{k-2}(x)$. k is the order of polynomials. Here, $\tilde{\mathbf{L}} = (2\mathbf{L}/\lambda_{\max}) - \mathbf{I}_N$ scales the Laplacian matrix $\mathbf{L}$ to $[-1, 1]$, where $\lambda_{\max}$ is the largest eigenvalue of $\mathbf{L}$.

To reduce computational complexity of the Laplacian matrix calculation in large-scale systems, we adopt first-order Chebyshev approximation, which localizes the convolution within immediate node neighborhoods:

$$\mathbf{g}_t^{(\ell)} \approx \theta_0^{(\ell)} \mathbf{X}_t^{(\ell)} + \theta_1^{(\ell)} \tilde{\mathbf{L}} \mathbf{X}_t^{(\ell)}. \quad (6)$$

The reparameterization $\theta_{l_0} = \theta_0 - \theta_1$ and $\theta_{l_1} = 2\theta_1/\lambda_{\max}$, combined with the parameter constraint $\theta_{l_0} = -\theta_{l_1}$ and $\lambda_{\max} \approx 2$ ($\lambda_{\max}$ typically ranges from 1.94 to 2.00 for power systems), leads to the simplified graph convolution form:

$$\mathbf{X}_t^{\text{Gconv},(\ell)} \approx \sigma_{\text{ReLU}} \left(\theta^{(\ell)} \left(\mathbf{I}_N + \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} \right) \mathbf{X}_t^{(\ell)} \right), \quad (7)$$

where $\theta^{(\ell)}$ serves as a single trainable parameter controlling the aggregation weights of neighboring node information. $\sigma_{\text{ReLU}}(\cdot)$ is the rectified linear unit (ReLU) function. $\mathbf{X}_t^{\text{Gconv},(\ell)} \in \mathbb{R}^{N \times C_S}$ is the output of graph convolution layer.

D. Physics-Informed Multi-Scale Adaptive Graph Learning

Since a single adjacency relationship in (7) cannot fully capture power system spatial dependencies, PI-MSA employs three distinct adjacency pathways from different physical perspectives, as illustrated Fig. 3.

Scale 1. Physical Grid Topology: First, the predefined physical grid topology matrix $\tilde{\mathbf{A}}^{\text{basic}} \in \mathbb{R}^{N \times N}$ constrains information flow strictly to directly connected local neighbors:

$$\mathbf{A}^{\text{basic}} = [a_{ij}]_{N \times N}, \quad \text{where } a_{ij} \in \{0, 1\}, \quad (8)$$

where $a_{ij} = 1$ if nodes i and j are connected. otherwise $a_{ij} = 0$.

Scale 2. Spatial-Temporal Adaptive Graph Learning: The system topology can shift due to contingencies, and effective distances between nodes may vary during events. To capture such time-varying dependencies, an adaptive adjacency relationship is introduced that learns from node features:

$$a_{ij,t}^{\text{adapt}} = \sigma_{\text{ReLU}}(w \cdot \text{sim}(\mathbf{x}_{i,t}, \mathbf{x}_{j,t}) + b), \quad (9)$$

$$\text{sim}(\mathbf{x}_{i,t}, \mathbf{x}_{j,t}) = \frac{\mathbf{x}_{i,t} \cdot (\mathbf{x}_{j,t})^\top}{\|\mathbf{x}_{i,t}\|_2 \cdot \|\mathbf{x}_{j,t}\|_2}, \quad (10)$$

where $\sigma_{\text{ReLU}}(\cdot)$ denotes the LeakyReLU activation function, $w, b \in \mathbb{R}$ are trainable weight and bias, respectively. $\mathbf{x}_{i,t}, \mathbf{x}_{j,t} \in \mathbb{R}^{C_S}$ and $\text{sim}(\mathbf{x}_{i,t}, \mathbf{x}_{j,t})$ computes similarity between nodes i and j at time t , $\|\cdot\|_2$ denotes the L2 norm. Thus, $\tilde{\mathbf{A}}^{\text{adapt}} = [a_{ij,t}^{\text{adapt}}]_{T \times N \times N}$ captures the instantaneous connection strength that evolve with system dynamics.

Scale 3. Physics-Informed Adaptive Graph Learning: Data-driven correlations of Scale 2 may yield physically inconsistent relationships, particularly for edge events whose signals attenuate along transmission lines and require consistent multi-sensor analysis. To address this challenge, a physics-informed adaptive adjacency matrix $\tilde{\mathbf{A}}^{\text{physics}}$ is constructed by combining shortest-path admittance encoding with adaptive learning $\tilde{\mathbf{A}}^{\text{adapt}}$. First, the shortest paths between all node pairs are computed using Dijkstra's algorithm on the distance-weighted adjacency matrix

$\tilde{\mathbf{A}}^{\text{distance}} = \tilde{\mathbf{A}}^{\text{basic}} \odot \mathcal{D}^{\text{line}}$, where $\mathcal{D}^{\text{line}} \in \mathbb{R}^{N \times N}$ stores the line distances:

$$\mathcal{SP} = \text{Dijkstra}(\tilde{\mathbf{A}}^{\text{distance}}). \quad (11)$$

For each node pair (i, j) , $\mathcal{SP}_{ij}$ stores the shortest path and $|\mathcal{SP}_{ij}|$ denotes the number of edges along this path. The admittance representation is then encoded as:

$$\mathbf{z}'_{ij} = \frac{1}{|\mathcal{SP}_{ij}|} \sum_{n=1}^{|\mathcal{SP}_{ij}|} \mathbf{z}_{ij,n} \Phi_{ij,n}^\top, \quad \mathbf{p}_{ij} = \frac{1}{\mathbf{z}'_{ij}}, \quad (12)$$

where $\mathbf{z}_{ij} = [R_1, \mathcal{X}_1, R_0, \mathcal{X}_0] \in \mathbb{R}^4$ contains the positive- and zero-sequence physical resistance $R_{1/0}$ and reactance $\mathcal{X}_{1/0}$ of the n -th edge, and $\Phi_{ij,n}^\top \in \mathbb{R}^{4 \times C_S}$ is a learnable projection matrix. The final shortest path admittance representation $\mathbf{p}_{ij} \in \mathbb{R}^{C_S}$ between nodes i and j is obtained by taking the reciprocal of encoded impedance representation $\mathbf{z}'_{ij} \in \mathbb{R}^{C_S}$. Finally, this encoded admittance vector $\mathbf{p}_{ij}$ explicitly gates the adaptive graph:

$$a_{ij}^{\text{P}} = \left(\sigma_{\text{ReLU}}(\beta) + a_{ij}^{\text{adapt}} \right) \mathbf{p}_{ij}, \quad (13)$$

where β is a learnable parameter and a_{ij}^{adapt} dynamically modulates the admittance encoding $\tilde{\mathbf{A}}^{\text{physics}} = [a_{ij}^{\text{P}}]_{T \times N \times N \times C_S}$. In practice, this formulation ensures strong connections are only formed between nodes exhibiting both high statistical data-correlation and tight physical electrical coupling.

E. Multi-Scale Graph Feature Fusion

The STMS-Block computation process can be expressed as:

$$\mathbf{X}^{\ell+1} = \mathcal{F}_\Gamma^{(\ell,2)} \left(\sigma_{\text{ReLU}} \left(\mathcal{G}_\theta^{(\ell)} \left(\mathcal{F}_\Gamma^{(\ell,1)}(\mathbf{X}^{(\ell)}) \right) \right) \right), \quad (14)$$

where $\mathbf{X}^{(\ell+1)} \in \mathbb{R}^{(T-2(K-1)) \times N \times C_T}$, and $\mathcal{F}_\Gamma^{(\ell,1)}(\cdot)$ and $\mathcal{F}_\Gamma^{(\ell,2)}(\cdot)$ denote the first and second temporal feature learning layers in the ℓ -th STMS-Block, $\mathcal{G}_\theta^{(\ell)}(\cdot)$ represents the multi-scale spatial graph feature learning layer. Then, the outputs from three scales are concatenated to form $\mathcal{H} = [\mathcal{H}_1 || \mathcal{H}_2 || \mathcal{H}_3] \in \mathbb{R}^{(T-2(K-1)) \times N \times 3C_T}$, and multi-head attention (MHA) is employed to adaptively fuse the graph features:

$$\text{MHA}(\mathcal{H}) = \left[\parallel_1^n \sigma_{\text{softmax}} \left(\frac{\mathbf{Q}_i(\mathbf{K}_i)^\top}{\sqrt{d_{\text{head}}}} \right) \mathbf{V}_i \right] \mathbf{W}^O, \quad (15)$$

where $\mathbf{Q}_i = \mathcal{H} \mathbf{W}_i^Q$, $\mathbf{K}_i = \mathcal{H} \mathbf{W}_i^K$, $\mathbf{V}_i = \mathcal{H} \mathbf{W}_i^V$ are query, key and value matrices, $\mathbf{W}_i^Q, \mathbf{W}_i^K, \mathbf{W}_i^V \in \mathbb{R}^{3C_T \times d_{\text{head}}}$ are learnable projection matrices, $\mathbf{W}^O \in \mathbb{R}^{(n \times d_{\text{head}}) \times C}$ is the output projection matrix. $\parallel_1^n$ represents matrix concatenation from head i to n .

F. Computational Complexity Analysis

The computational complexity of PI-MSA is analyzed component-wise. Let N denote the number of nodes, C the feature dimension, T the temporal length, and $|\mathcal{E}|$ the number of edges. 1. TGC Layer has complexity: $\mathcal{O}_{\text{TGC}}(NC^2T)$; 2. The three parallel graph convolution layers have: 1) Grid Topology: $\mathcal{O}(|\mathcal{E}|CT + NC^2T) \approx \mathcal{O}_{\text{Grid}}(NC^2T)$ since $|\mathcal{E}| \approx \mathcal{O}(N)$ for sparse power grids; 2) Adaptive: $\mathcal{O}_{\text{adapt}}(NC^2T)$ using $T_s = 6(T_s \ll T)$ sampled points with top-4 neighbors; 3) Physics-informed: $\mathcal{O}_{\text{PI}}(NC^2T)$ by reusing the adaptive adjacency with

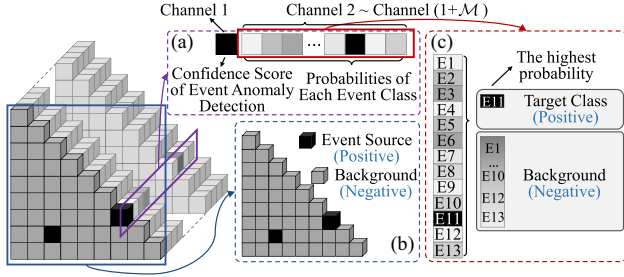


Fig. 4. Spatial-aware and task-coupled 3-D output structure for unified ED-L-C task.

$\mathcal{O}(N)$ additional cost. 3. Multi-scale fusion: MHA on $N \times 3C \times d$ features has complexity $\mathcal{O}_{\text{MHA}}(NCd^2)$. Therefore, the overall complexity is $\mathcal{O}_{\text{PL-MSA}} = \mathcal{O}_{\text{STMS}} + \mathcal{O}_{\text{MHA}} \approx \mathcal{O}_{\text{PL-MSA}}(NC^2T)$ when $C \approx d \ll T$. The model scales approximately linearly with N , making it scalable to grids ranging from dozens to thousands of nodes.

IV. TRAINING AND INFERENCE PHASE FOR ED-L-C TASK

A. Multi-Source ED-L-C Task-Aware Loss Design

The model outputs a 3-D tensor of shape $[N, N, 1+\mathcal{M}]$ that extends the adjacency matrix concept, as shown in Fig. 4. Diagonal elements (i, i) represent node events, off-diagonal elements (i, j) represent edge events, and only the lower triangular part is computed due to graph symmetry. The first channel [gray, Fig. 4(b)] represents anomaly detection confidence scores, while subsequent channels [lighter shades, Fig. 4(c)] indicate event class probabilities. As illustrated in Fig. 4(a), task dependencies are embedded explicitly in the label design: when event class ω occurs at node i , both Channel 1 (detection) and Channel $(1+\omega)$ (classification) at position (i, i) are labeled as 1, thereby coupling detection confidence and classification probability within the same structure during training. This design enables the model to localize events spatially through tensor indices, while the predicted event class is determined by ranking probabilities across all channels, with classification conditioned on detection confidence. The loss function is designed to balance the contributions of positive and negative examples:

$$\mathcal{L}_{\text{total}} = \alpha_1 \mathcal{L}_{\text{pos}}^{\text{anomaly}} + \alpha_2 \mathcal{L}_{\text{neg}}^{\text{anomaly}} + \gamma_1 \mathcal{L}_{\text{pos}}^{\text{class}} + \gamma_2 \mathcal{L}_{\text{neg}}^{\text{class}}, \quad (16)$$

where the anomaly detection loss is computed as:

$$\mathcal{L}_{\text{pos/neg}}^{\text{anomaly}} = \sum_{i=0}^{N-1} \sum_{j=0}^i \mathbb{I}_{ij}^{\text{pos/neg}} (y_{ij} - \hat{y}_{ij})^2, \quad (17)$$

and the event classification loss is defined as:

$$\mathcal{L}_{\text{pos/neg}}^{\text{class}} = \sum_{i=0}^{N-1} \sum_{j=0}^i \mathbb{I}_{ij}^{\text{pos/neg}} \sum_{\omega=0}^{\mathcal{M}} (\rho_{ij,\omega} - \hat{\rho}_{ij,\omega})^2. \quad (18)$$

Here, $\mathbb{I}_{ij}^{\text{pos/neg}}$ denotes the indicator function for positive/negative examples, $\mathbb{I}_{ij}^{\text{pos}} = 1$ for nodes/edges with events, $\mathbb{I}_{ij}^{\text{neg}} = 1$ for nodes/edges without events. y_{ij} and $\hat{y}_{ij}$ are the predicted anomaly scores and ground truth between node i and j , respectively, while $\rho_{ij,\omega}$ and $\hat{\rho}_{ij,\omega}$ correspond to the predicted

TABLE II
TEST SYSTEMS SUMMARY

Test System	Voltage Level	System Type	Key Characteristics	Included Events
IEEE 34-node	24.9 kV	Distribution	Actual Arizona grid	E1–E13
IEEE 123-node	4.16 kV	Distribution	Complex topology with multiple switching devices	E1–E13
New England 39-node	345 kV	Transmission	Meshed topology with synchronous generators	E1–E10, E17–E19
European 906-node	Mixed	Distribution	Large-scale network for scalability evaluation	E1–E10
Modified IEEE 33-node	12.66 kV	Distribution + RES	Integrated Photovoltaic (PV), wind turbine, and BESS	E1–E16

TABLE III
EVENT CLASS DEFINITIONS

Event ID	Description	Location
E1/E2/E3	Single-line-to-ground (A-G / B-G / C-G)	Node and Edge events
E4/E5/E6	Line-to-line (A-B / B-C / C-A)	
E7/E8/E9	Two phase line-to-ground (AB-G / BC-G / AC-G)	
E10	Three-phase line-to-ground (ABC-G)	Node events
E11/E12/E13	Three-phase Motor Load switching / UPS Load switching / Capacitor Bank switching	
E14/E15/E16	Wind Turbine / PV System / BESS Disconnection	Node events
E17/E18	Generator Trip / Load Loss	Node events
E19	Line Trip	Edge events

probabilities and ground truth for event class ω between node i and j . The hyperparameters α_1 , α_2 , γ_1 , and γ_2 balance the contributions of positive and negative examples in both detection and classification tasks, preventing the loss computation from being dominated by background negative examples.

B. Reasoning Process for Multi-Source ED-L-C Task

For inference phase, the detection channel serves as a gating mechanism that filters out locations with low confidence scores, and classification is performed only on the remaining candidates. This explicitly models the conditional probability chain in a single forward pass. The final prediction score for each potential location (i, j) is computed as:

$$\text{score}_{i,j} = \underbrace{\mathbb{I}(y_{ij} > \tau)}_{\mathbb{I}_{ij}^{\text{pos/neg}}} \times y_{ij} \times \max_{\omega \in 1, \dots, \mathcal{M}} \underbrace{\text{Pr}(\text{cls}_{\omega} | \text{event})_{i,j}}_{\rho_{ij,\omega}}, \quad (19)$$

where $\text{Pr}(\text{cls}_{\omega} | \text{event})$ represents the conditional probability distribution over $\mathcal{M}$ event classes, and $\mathbb{I}(y_{ij} > \tau)$ denotes locations with confidence scores below threshold τ are filtered out.

V. EXPERIMENTAL SETUP

A. Test Power Systems and Event Scenarios

The proposed framework is evaluated on five IEEE benchmark systems spanning different scales, voltage levels, and topological characteristics to comprehensively assess its generalization capability. Table II summarizes the test systems and their corresponding event configurations. The node numbering and system configurations are consistent with [32].

The event class (E1–E19) included in Table II are defined in Table III, covering fault events, harmonic load switching events, RES-related events, and transmission-specific events at both node and edge levels.

B. Monte Carlo Based Multi-Source Events Generation

The multi-source events generation process is demonstrated using the IEEE 34-node system as an example. The Monte Carlo method is employed to generate 13 different classes of events. The detailed generation procedure is presented in Algorithm 1.

All measurements are recorded at 120 Hz with a 1-s sliding window. As the power system operates, samples are updated through sliding windows. Each sample represents a graph time-series data $\mathbf{X} \in \mathbb{R}^{N \times C \times T}$, where unmeasured nodes and phases are zero-padded to indicate the absence of active electrical activity. The historical data samples are manually annotated with labels in the format described in Section IV-A. Unless otherwise specified, all experiments assume full observability. Each system supports up to 8 event sources per observation window, except for the 906-node system which supports up to 16 to further evaluate scalability. Each dataset contains 4000–8000 event samples and 400–1000 normal samples.

C. Implementation Details

The dataset is split chronologically into training (70%), validation (10%), and test (20%) sets to prevent information leakage, with training samples shuffled before feeding to the model. The model checkpoint with the best performance on the validation set is saved for evaluation. The proposed model contains 2 STMS-Block, where spatial and temporal kernel sizes are set to 3, with 64 temporal convolution kernels and 16 spatial graph convolution kernels. The final temporal learning layer employs convolution kernels of length 3, followed by a 3-head MHA layer with one head per pathway, fusing the features learned across the three pathways, and a stack of linear layers mapping the fused representations to the target ED-L-C output tensor.

The model is trained using Adam optimizer (weight decay $1.0e-4$) with a MultiStepLR learning rate scheduler. The learning rate starts at 0.002 and decreases by half at epochs [1, 50, 90]. Gradient clipping (3.0) and dropout regularization (rate 0.5) are employed to ensure stable training. The loss function $\alpha_1, \alpha_2, \gamma_1$, and γ_2 are set to [1.0, 0.5, 1.0, 0.7]. The model is trained for 100 epochs with a batch size of 64. All hyperparameters were determined through empirical and experimental tuning.

D. Baseline Methods and Evaluation Metrics

Evaluation Metrics: To comprehensively evaluate the model performance, this study adopts Correct Localization (CorLoc), Average Precision (AP) and mean Average Precision (mAP) metrics. The CorLoc is defined as the percentage of system events that are correctly localized:

$$\text{CorLoc} = \frac{1}{E_{\text{num}}} \sum_{i=1}^{E_{\text{num}}} \mathbb{I}(\text{Loc}_i = \text{gtLoc}_i), \quad (20)$$

where E_{num} is the total number of detected events, $\mathbb{I}(\cdot)$ denotes the indicator function that equals 1 when predicted localization Loc_i matches the ground truth gtLoc_i , and 0 otherwise. For joint evaluation of ED-L-C performance, True Positive (TP) is defined

Algorithm 1: Monte Carlo Randomized Multi-Source Event Data Generation.

Input: Stable system operation starting at time t_{start}

Output: Multi-source event graph time-series dataset

1. Sample $t_{\text{interval}} \sim \text{Uniform}[0.5, 1.0]$ seconds
2. Set event generation interval: $[t_{\text{start}}, t_{\text{end}}]$, where $t_{\text{end}} = t_{\text{start}} + t_{\text{interval}}$
3. Sample $E_{\text{num}} \sim \text{Uniform}\{1, 2, 3, 4, 5, 6, 7, 8\}$
4. Initialize candidate location set E_{loc}
5. Sample E_{num} locations from E_{loc} without replacement
6. **For** each edge event location **do**
7. Sample $\alpha_{\text{line}} \sim \text{Uniform}(0, 1)$
8. Set position at $\alpha_{\text{line}} \cdot L$ (L =line length)
9. **End for**
10. **For** each selected location $E^{(i)}$ **do**
11. Randomly select event class from event set: $E_{\text{cls}} = \{E_1, \dots, E_{13}\}$
12. **if** $E^{(i)}$ is a fault event **then**
13. Sample $R_{\text{base}} \in \{0.01, 0.1, 1, 5, 15, 20\}\Omega$, $\alpha_{\text{fault}} \sim \text{Uniform}[0.8, 1.2]$
14. Compute $R_{\text{fault}} = R_{\text{base}} \times \alpha_{\text{fault}}$
15. **End if**
16. Sample $t_{\text{trigger}}^{(i)} \sim \text{Uniform}(t_{\text{start}}, t_{\text{end}})$ and $\alpha_{\text{duration}} \sim \text{Uniform}(-1, 1)$
17. Calculate $t_{\text{end}}^{(i)} = t_{\text{start}} + t_{\text{trigger}}^{(i)} + \Delta t^{(i)}$, $\Delta t^{(i)} = (3 + \alpha_{\text{duration}}) / f_{\text{system}}$
18. **End for**

Return: Randomized graph time-series events data

based on both event classification and topological distance:

$$\text{TP} = \{\text{pred}_i | \mathbb{I}(\text{cls}_i = \text{gt}_i) \times \mathbb{I}(\text{D}_{n\text{-hop}}(\text{Loc}_i, \text{gtLoc}_i))\}. \quad (21)$$

When calculating AP, we require exact location matching where: $\mathbb{I}(\text{D}_{0\text{-hop}}(\text{Loc}_i, \text{gtLoc}_i)) \rightarrow \mathbb{I}(\text{Loc}_i = \text{gtLoc}_i)$. The AP of each event class and mAP across all classes are defined as:

$$\text{mAP} = \frac{1}{\mathcal{M}} \sum_{\omega=1}^{\mathcal{M}} \text{AP}_{\omega}, \text{AP} = \int_0^1 \text{P}(\text{R}) \text{dR}, \quad (22)$$

where $\text{P}(\text{R})$ represents the precision $\text{P} = \text{TP}/(\text{TP}+\text{FP})$ at recall level $\text{R} = \text{TP}/(\text{TP}+\text{FN})$.

Furthermore, three topology-aware metrics are introduced: mean Average Precision for node events (mAP_N), mean Average Precision for edge events (mAP_E), and One-hop mean Average Precision (mAP_{OH}). The mAP_{OH} relaxes the exact matching requirement by accepting predictions within one-hop of the ground truth $\mathbb{I}(\text{D}_{1\text{-hop}}(\text{Loc}_i, \text{gtLoc}_i))$, allowing us to assess ED-L-C performance on approximate event localization.

Baselines Models: For fair comparison, we select a wealth of baseline models with publicly available official implementations [33] that have been validated for spatial-temporal modeling, including conventional time series models (LSTM, MLP(Linear), and NBeats), classic graph-based models (STGCN, DCRNN, Graph WaveNet, and AGCRN), and recent state-of-the-art time

TABLE IV
COMPARISON OF DIFFERENT METHODS ON MULTI-SOURCE ED-L-C TASK PERFORMANCE OF 34-NODE AND 123-NODE SYSTEMS

Metrics		IEEE 34-node test system					IEEE 123-node test system				
Methods		mAP(%)	CorLoc(%)	mAP _{OH} (%)	mAP _E (%)	mAP _N (%)	mAP(%)	CorLoc(%)	mAP _{OH} (%)	mAP _E (%)	mAP _N (%)
Model Backbone	STGCN ⁽¹⁾	75.37	87.80	81.65	70.11	90.68	74.89	90.71	75.77	69.53	90.07
	DCRNN ⁽¹⁾	69.33	75.57	75.98	62.25	90.42	76.33	90.28	76.37	73.85	85.70
	Graph WaveNet ⁽¹⁾	74.97	88.52	79.04	71.06	87.30	75.99	90.76	78.00	72.04	89.13
	AGCRN ⁽¹⁾	76.19	88.29	78.97	70.29	93.95	73.57	90.31	75.39	70.81	84.21
	PatchTST ⁽²⁾	63.11	72.12	63.79	54.22	90.01	66.13	88.57	67.56	65.57	69.75
	iTransformer ⁽²⁾	69.07	75.39	69.30	62.95	88.94	69.48	91.32	69.60	70.38	67.54
	Crossformer ⁽²⁾	67.35	73.26	68.93	62.53	81.90	70.31	92.57	71.84	65.34	87.29
	TimesNet ⁽²⁾	75.27	87.51	81.52	72.02	83.81	80.22	89.78	82.18	79.44	84.19
	NBeats ⁽²⁾	75.71	84.70	77.57	73.00	84.69	79.03	90.84	79.91	78.02	84.23
	MLP ⁽²⁾	29.82	86.58	31.76	11.38	89.04	30.66	86.64	32.32	22.42	61.56
	LSTM ⁽²⁾	43.65	72.01	45.86	30.47	87.49	43.65	87.31	46.40	47.63	33.55
	PI-MSA	81.21	90.18	85.49	76.50	95.36	84.62	92.16	89.68	83.62	90.31

In the table, ⁽¹⁾ means graph-based models, and ⁽²⁾ is non-graph-based models. Bold: best (black), second-best (orange), third-best (blue) results.

series models (PatchTST, iTransformer, TimesNet, and Crossformer). We only added a uniform ED-L-C prediction head to each model without any other modifications. It is worth noting that ED-L-C tasks require end-to-end fast processing capabilities, as discussed in Section IV. Hence, we focus on deep learning baselines readily adaptable to our unified ED-L-C task formulation. All models were developed with Pytorch, PyG, and EasyTorch, conducting training on an NVIDIA A40 48 GB GPU for 100 epochs.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

This section evaluates the proposed PI-MSA framework in four parts: 1) overall performance evaluation on power grid of varying scales, 2) robustness analysis under challenging conditions including reduced observability, topology variations, and imperfect data quality, 3) generalization validation on transmission and RES-integrated systems, and 4) model analysis including ablation studies and computational efficiency.

A. Overall ED-L-C Performance Evaluation

1) *Baseline Comparison*: Table IV compares ED-L-C performance on the 34-node and 123-node systems. PI-MSA achieves 81.21% and 84.62% mAP on the two systems, exceeding the best baselines by 5.02% and 4.4%, respectively. This consistent improvement is attributed to the physics-aligned multi-scale architecture of PI-MSA, which facilitates feature extraction across local-to-global topology.

The comparison also reveals a clear performance hierarchy: Most graph-based models outperform non-graph-based models while requiring fewer parameters, as explicit physical topology provides valuable structural priors for joint node and edge event analysis. In contrast, Transformer-based models struggle to resolve overlapping waveform patterns from spatially distributed sources without explicit structural guidance. Basic models (MLP, LSTM) perform poorly overall, as they lack the spatial-temporal synergy essential for multi-source ED-L-C.

Notably, most models exhibit lower mAP_E than mAP_N, with performance gaps primarily concentrated in edge events. This is because node events captured by co-located sensors with higher signal fidelity, and certain harmonic load switching events exhibit distinctive signatures that facilitate recognition.

2) Performance Under Different Number of Event Sources:

To evaluate PI-MSA under multi-source scenarios, trained models are tested on eight datasets containing 1–8 event sources, with results presented in Fig. 5. For (1-2)-sources events (Fig. 5(a1)), the model performs better on the 34-node system due to simpler topology. For 3–8 sources (Fig. 5 (a/b 2-4)), the 123-node system yields superior results, as its larger spatial domain better separates overlapping propagation patterns. Specifically, the model maintains CorLoc around 92% across all scenarios with mAP above 79% even under eight concurrent events (Fig. 5(b4)), attributed to multi-scale graph learning with MHA that effectively discriminates wave propagation nodes from event source nodes under increased spatial-temporal dependencies.

3) *Scalability to Large-Scale Power System*: To further validate scalability, we evaluate PI-MSA on the European 906-node system, increasing the maximum number of event sources from 8 to 16 per observation window, with 16.1% of test events occurring at previously unseen locations. PI-MSA achieves 96.42% mAP and 98.79% CorLoc overall. As shown in Fig. 6, per-class AP across seen and unseen locations shows a gap of only 0.23% (96.26% vs. 96.03% mAP), demonstrating effective spatial generalization stemming from the unified ED-L-C training objective and adaptive graph learning, which guide the model to learn transferable structural patterns rather than location-specific mappings.

B. Robustness Analysis of PI-MSA

1) *Robustness Under Reduced Observability*: Fig. 7 illustrates ED-L-C performance as D-PMU coverage decreases from 100% to 12% on the 34-node system. Baseline methods like DCRNN and STGCN that rely on static graph structures exhibit degradation exceeding 40%, as sparse sensor deployment disrupts message passing paths along the physical topology. In contrast, PI-MSA maintains 54.45% mAP and 73.13% CorLoc at minimal 12% coverage (adequate for operational monitoring). This robustness stems from the multi-scale architecture integrating adaptive and shortest-path admittance-based physics-informed graph learning, which provides alternative message passing paths based on feature similarity and electrical distance when observed nodes are physically non-adjacent.

2) *Robustness Under Topology Variations*: We evaluate robustness under high-frequency network reconfiguration on the

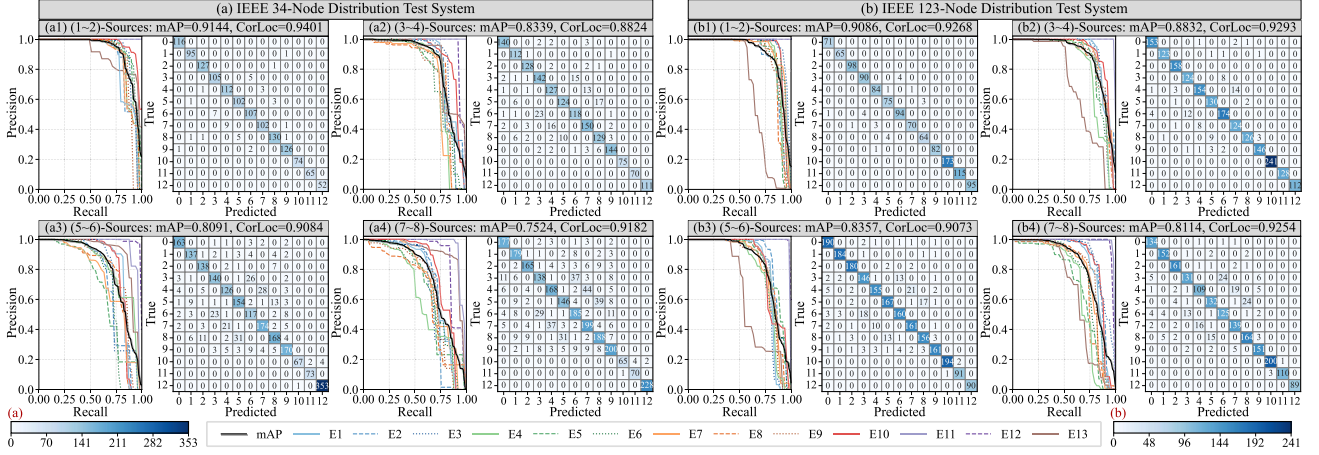


Fig. 5. AP and confusion matrices of PI-MSA under different number of event source scenarios. (a) IEEE 34-node system. (b) IEEE 123-node system. In each 2×2 panel, subfigures (1)–(4) correspond to scenarios with 1–2, 3–4, 5–6, and 7–8 event sources, respectively.

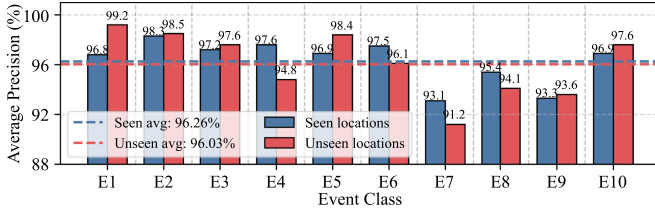


Fig. 6. Per-class AP on seen and unseen locations.

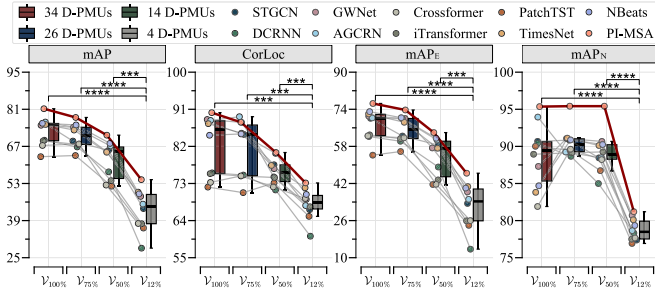


Fig. 7. Impact of different number of sensors on ED-L-C task performance across different methods.

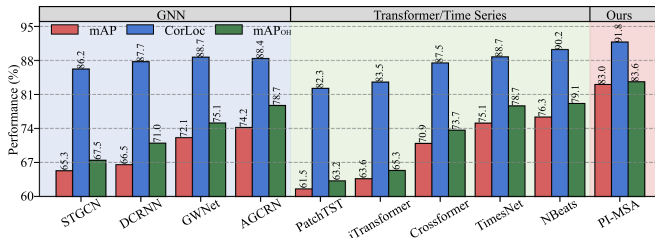


Fig. 8. Performance comparison under dynamic topology conditions.

123-node system, where five three-phase breakers at node pairs (60–160), (197–97), (151–300), (13–152), and (54–94) switch states at random intervals of 33–67 ms. From 2,000 generated samples, 1,000 are combined with static topology data from Section VI-A 1) for training and 1000 for testing, with a predefined adjacency matrix used throughout to simulate undetected topology changes. As shown in Fig. 8, PI-MSA achieves over 6.5%

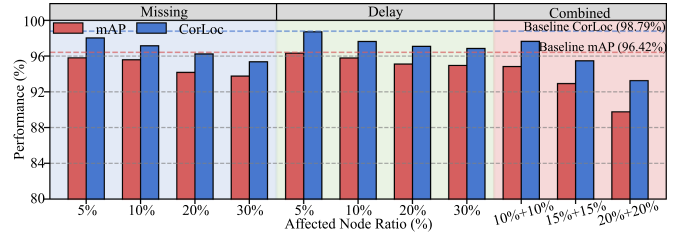


Fig. 9. Robustness to Missing and Delayed D-PMU Measurements.

higher mAP than the best baseline and maintains CorLoc above 91%, attributed to the adaptive graph learning mechanism that dynamically adjusts inter-node connection strengths regardless of the pre-defined physical topology.

3) Robustness to Missing and Delayed Measurements: We evaluate PI-MSA under missing data, delayed measurements, and their combination on the 906-node system. Missing data scenarios apply segment-based dropout (10–60 time steps, ~83–500 ms) to randomly selected D-PMU nodes; delayed nodes hold the last valid measurement and progressively recover to real-time flow. As shown in Fig. 9, under a 30% ratio, mAP drops by only 2.66% for missing data (96.42%→93.76%) and 1.47% for delays, the latter preserving temporal continuity. Under 20% missing and 20% delay combined, the model maintains 89.76% mAP with CorLoc above 93%. This robustness stems from the spatial-temporal adaptive graph learning mechanism, which dynamically leverages unaffected neighboring nodes to compensate for corrupted measurements.

C. Generalization to Different Grid Configurations

1) Generalization to Meshed Transmission Grid: As shown in Fig. 10, the proposed method achieves 86.13% mAP on the meshed high-voltage 39-node transmission system, with fault events (E1–E10) averaging 85.48% and operational events (E17–E19) averaging 87.94%. In particular, Generator Trip and Line Trip events achieve 97.89% and 90.16% AP respectively, the latter despite introducing dynamic topology changes.

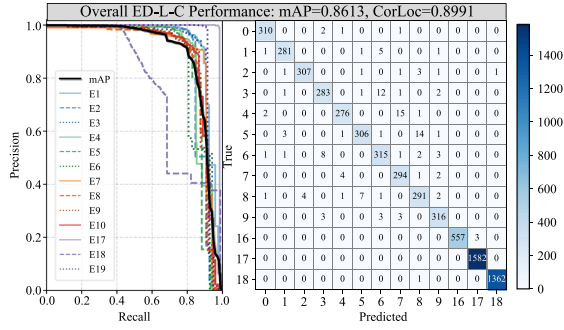


Fig. 10. Overall ED-L-C performance of PI-MSA on the New England 39-node transmission system.

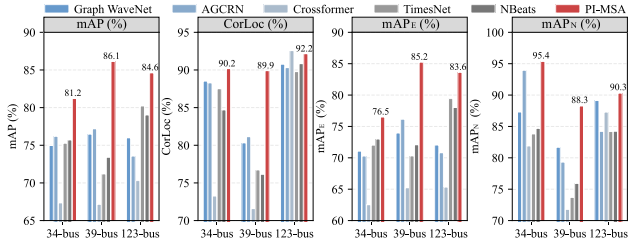


Fig. 11. Comparison with baseline methods across different systems.

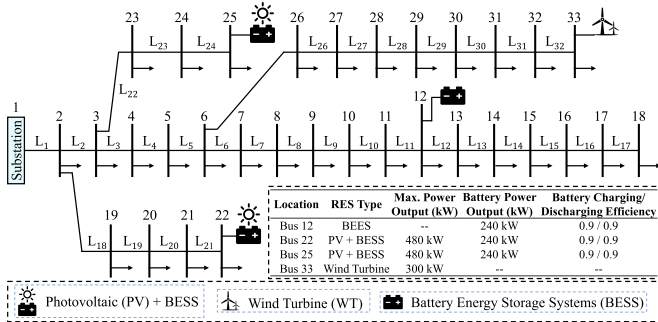


Fig. 12. Modified IEEE 33-node system with RES integration.

As shown in Fig. 11, PI-MSA consistently outperforms all baselines across distribution and transmission systems, exceeding the best baseline by over 8.5% in mAP. Time-series models (e.g., TimesNet and NBeats) exhibit pronounced degradation on meshed transmission grids, while graph-based methods with adaptive learning (e.g., AGCRN and Graph WaveNet) remain relatively stable. This is attributed to two factors: the physics-informed design encodes shortest-path impedance priors, organizing meshed multi-path connectivity into physically meaningful representations with adaptive topology handling; and the unified ED-L-C formulation jointly optimizes detection, localization, and classification under a single structured objective, enabling precise events pinpointing under complex and dynamic grid conditions.

2) Generalization to RES-Integrated Power Grid: The modified IEEE 33-node system is integrated with RES under PQ control mode, as illustrated in Fig. 12. RES integration introduces increased waveform complexity, including cross-phase disturbances and noticeable harmonic distortion.

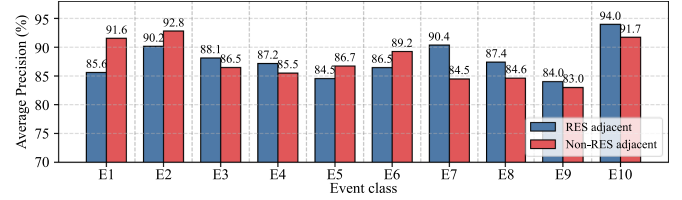


Fig. 13. Comparison of mAP and CorLoc for events E1-E10 at RES-adjacent and non-RES-adjacent locations.

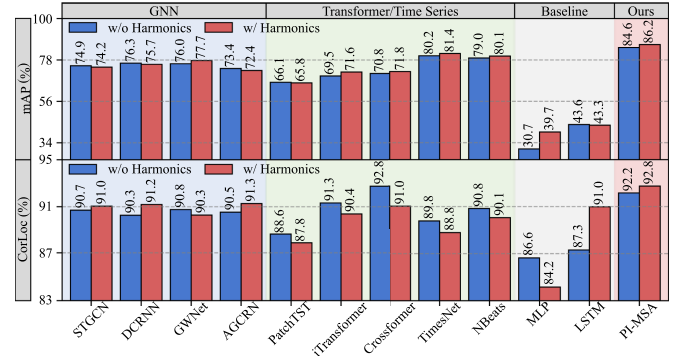


Fig. 14. ED-L-C performance comparison between without (w/o) harmonics and with (w) harmonics on ED-L-C tasks.

Despite these challenges, PI-MSA achieves 85.02% mAP and 90.18% CorLoc on this system. As shown in Fig. 13, events at RES-adjacent and non-RES-adjacent locations achieve 87.52% and 86.72% mAP respectively, a gap of less than 1%. Together with 80.67% mAP for RES-disconnection events, this confirms that multi-scale adaptive learning effectively extracts event signatures invariant to RES-induced interference.

D. Model Analysis

1) Impact of Harmonic Measurement (H-PMU): To investigate the impact of harmonic information on ED-L-C performance, we extend experiments to include third and fifth harmonic components on the 123-node system. As shown in Fig. 14, PI-MSA achieves a 1.6% mAP improvement and maintaining the highest performance among all methods. While most methods benefit from harmonic features that capture distinctive spatial-temporal event patterns, some models (e.g., AGCRN and PatchTST) show marginal or no improvement, indicating difficulty in exploiting high-dimensional harmonic features. This contrast highlights the advantage of PI-MSA in effectively leveraging harmonic information.

2) Ablation Studies: To validate the effectiveness of each component in PI-MSA, we conduct ablation studies on the IEEE 123-node system, including single-scale, dual-scale, and component-removal settings. As shown in Fig. 15(a), *Adaptive* learning achieves the best single-scale performance (81.33% mAP), followed by *Physics* (78.93%) and *Grid Topology* (75.62%). The *Physics* scale outperforms *Grid Topology* by 3.3%, as admittance encoding provides finer-grained electrical distance than binary connectivity. Fig. 15(b) shows that *Adaptive+Physics* yields the best dual-scale result (83.71% mAP),

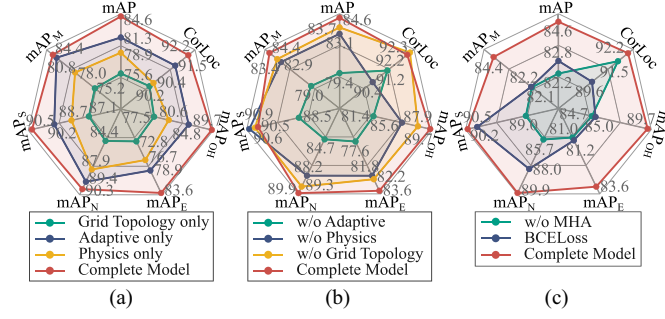


Fig. 15. Ablation study results for different modules of PI-MSA. (a) Single-scale. (b) Dual-scale combination. (c) Component analysis with key modules removed.

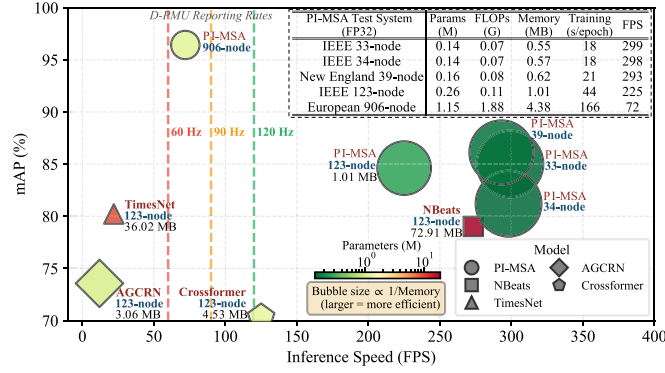


Fig. 16. Computational efficiency (runtime, memory, and ED-L-C performance) comparison across methods and system scales.

with 2.6% mAP_E gain over *Adaptive-only* in multi-source scenarios. The complete model further improves to 84.62% mAP . As shown in Fig. 15(c), removing MHA degrades performance, and the proposed loss function outperforms BCE by 2.3% by better balancing event and background examples.

3) Computational Efficiency: Key Baseline Comparison and Scalability Analysis: Fig. 16 demonstrates the computational efficiency and scalability of PI-MSA. As system size increases from 33 to 906 nodes, memory consumption grows modestly from 0.55 MB to 4.38 MB while maintaining real-time inference speeds (72–299 FPS), benefiting from the compact STMS-Block design with efficient convolutional operations. Compared with top-performing baselines on the 123-node system, NBeats achieves high speed but at the cost of significantly more parameters and memory. In contrast, PI-MSA requires only 0.26M parameters and 1.01 MB memory while achieving superior ED-L-C performance. This lightweight design enables an inference speed of 225 FPS, supporting real-time online event analysis across different D-PMU reporting rates (30–120 Hz).

VII. CONCLUSION

This study proposed a novel PI-MSA graph neural network for power system event analysis, unifying ED-L-C into an end-to-end framework that naturally supports multi-source event analysis without predefined event combination patterns, simultaneously performing node, and edge-level event analysis in one-shot inference. The physics-informed graph learning

guided by relative electrical distance enhances spatial feature learning, while the multi-scale architecture captures spatial-temporal event dynamics across different resolutions. Experimental results on systems with different voltage levels and grid topologies demonstrate that PI-MSA achieves over 5% mAP improvement compared with state-of-the-art methods. The model exhibits strong generalization and robustness, with less than 1% performance gap across seen versus unseen locations and RES-adjacent versus non-adjacent scenarios, while maintaining real-time inference at D-PMU reporting rates up to 120 Hz. Furthermore, incorporating third and fifth harmonic phasors yields an additional 1.6% mAP improvement, demonstrating the capability of proposed model to leverage richer frequency-domain signatures for enhanced event analysis.

The proposed unified ED-L-C formulation features a simple yet effective design that can be easily transferred to different deep learning backbones, enabling practitioners to leverage future advances in graph learning or time-series modeling. Moreover, the ED-L-C framework extends naturally to other graph-structured industrial event analysis scenarios, such as transportation incidents and cyber security incidents.

Despite these advantages, the current framework shows performance degradation when events are spatially adjacent and temporally highly coupled. Therefore, the future work will focus on: 1) refining the ED-L-C output structure with higher spatial resolution for dense event scenarios in compact grids, and 2) extending the framework to the cyber layer for ED-L-C of communication events and cyber-attacks, thereby further enhancing power system security and stability.

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